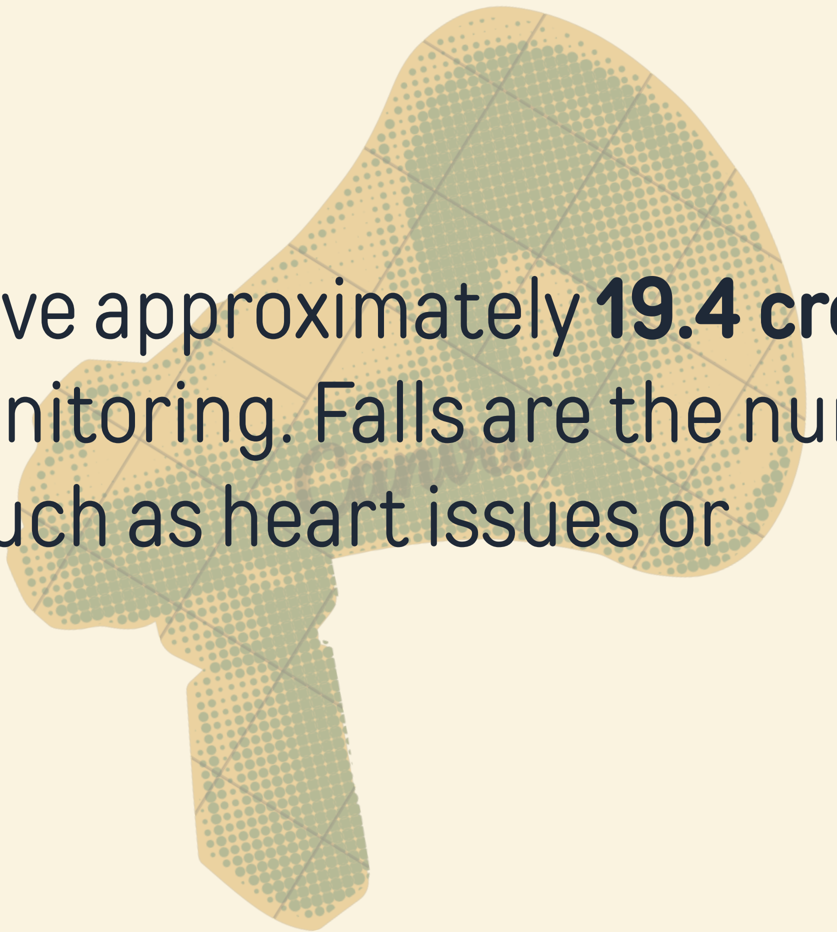
SPARS

A WiFi based system that detects human breathing , heart rate, pose of the person and presence without cameras or wearables.

Problem Statement



India (and globally) is facing a rapidly aging population. By 2031, India will have approximately **19.4 crore** elderly people (aged 60+), and over **70%** of them live without continuous monitoring. Falls are the number one cause of injury-related deaths among the elderly, and in emergencies such as heart issues or breathing problems, even a **small delay in detection can be fatal**.

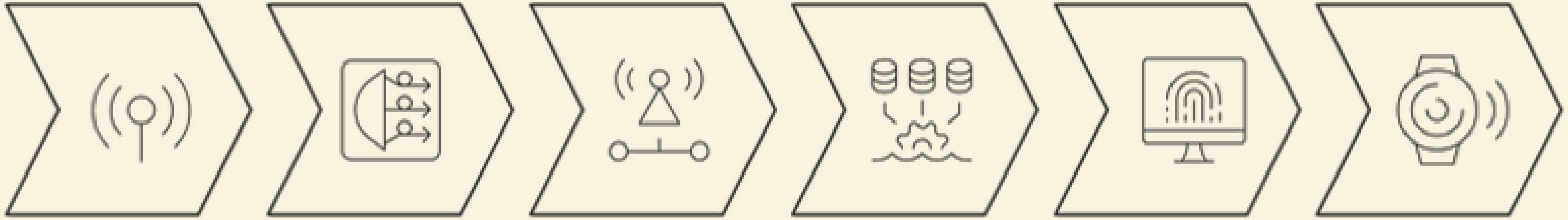
Why Current Solutions **FAIL**:

Solution	Problem
Cameras	Privacy invasion
Wearables	People forget / remove
Nurses / Manual care	Not scalable / Expensive

When every second matters, we still lack a simple, affordable, and privacy-safe way to protect our elders in real time.

Problem Solution

We use **WiFi signals** combined with **ESP32-S3** based electronics to detect human presence, breathing patterns, heart rate, motion, all without relying on cameras or wearable devices. The system works seamlessly through walls and remains effective even in challenging conditions such as darkness, smoke, or fire. By leveraging existing **WiFi infrastructure**, it enables continuous, real-time monitoring while being completely passive, requiring no user interaction, thereby ensuring both convenience and privacy.



WiFi Signals Transmitted
WiFi signals are broadcast into the environment

Signals Reflect
Signals bounce off human bodies and surroundings

CSI Captured
ESP32-S3 collects CSI from reflected signals

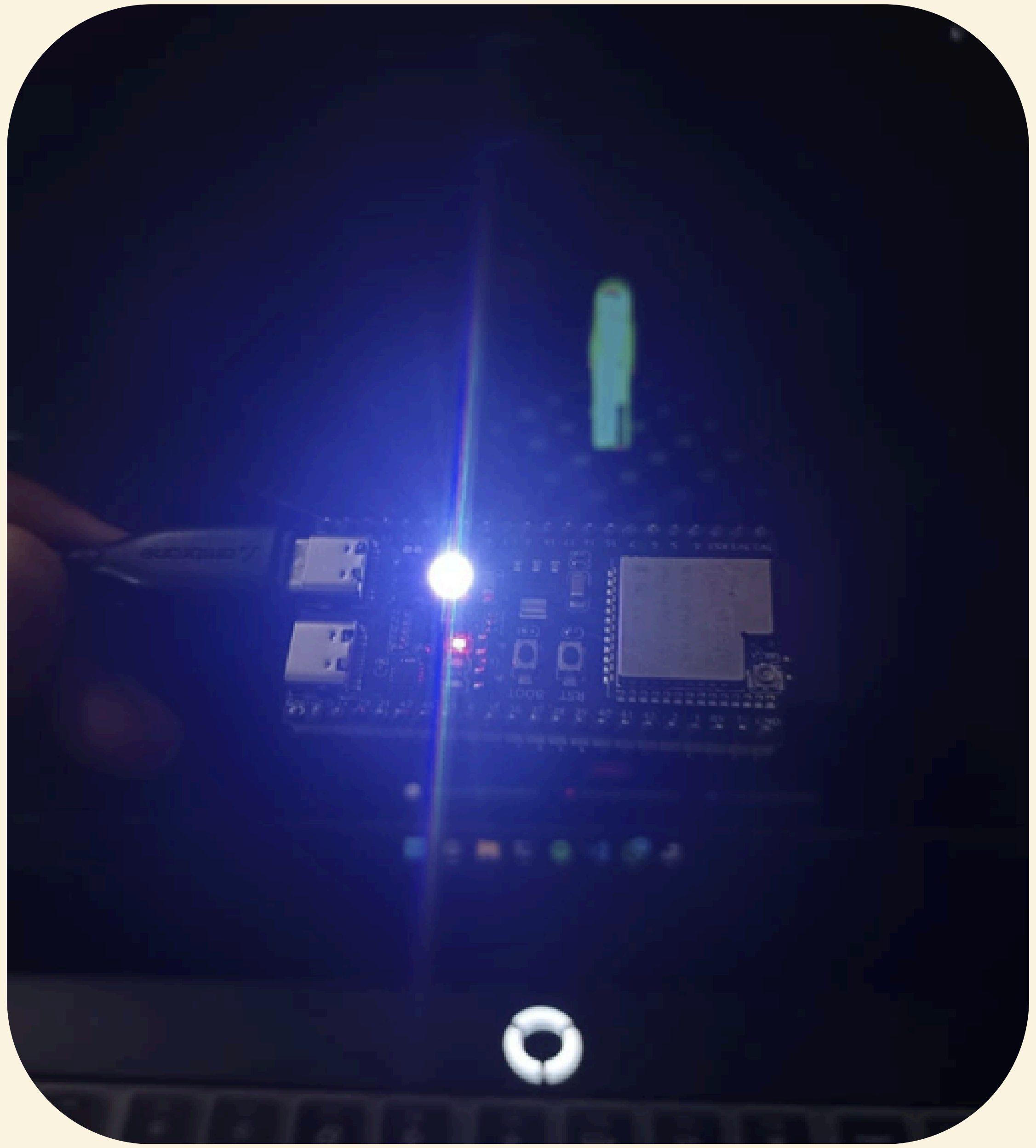
Data Processed
Algorithms analyze the collected data

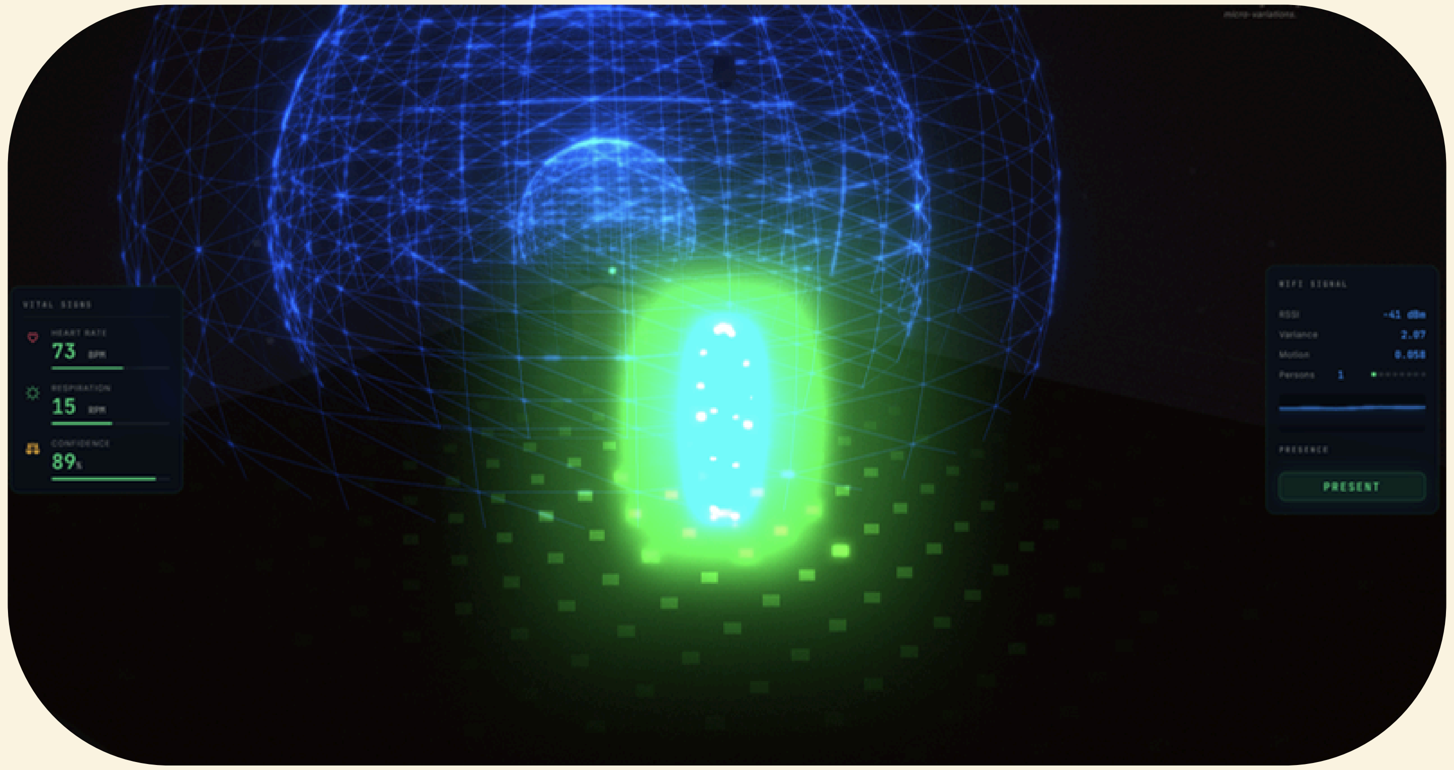
Patterns Analyzed
System identifies presence, motion, and breathing

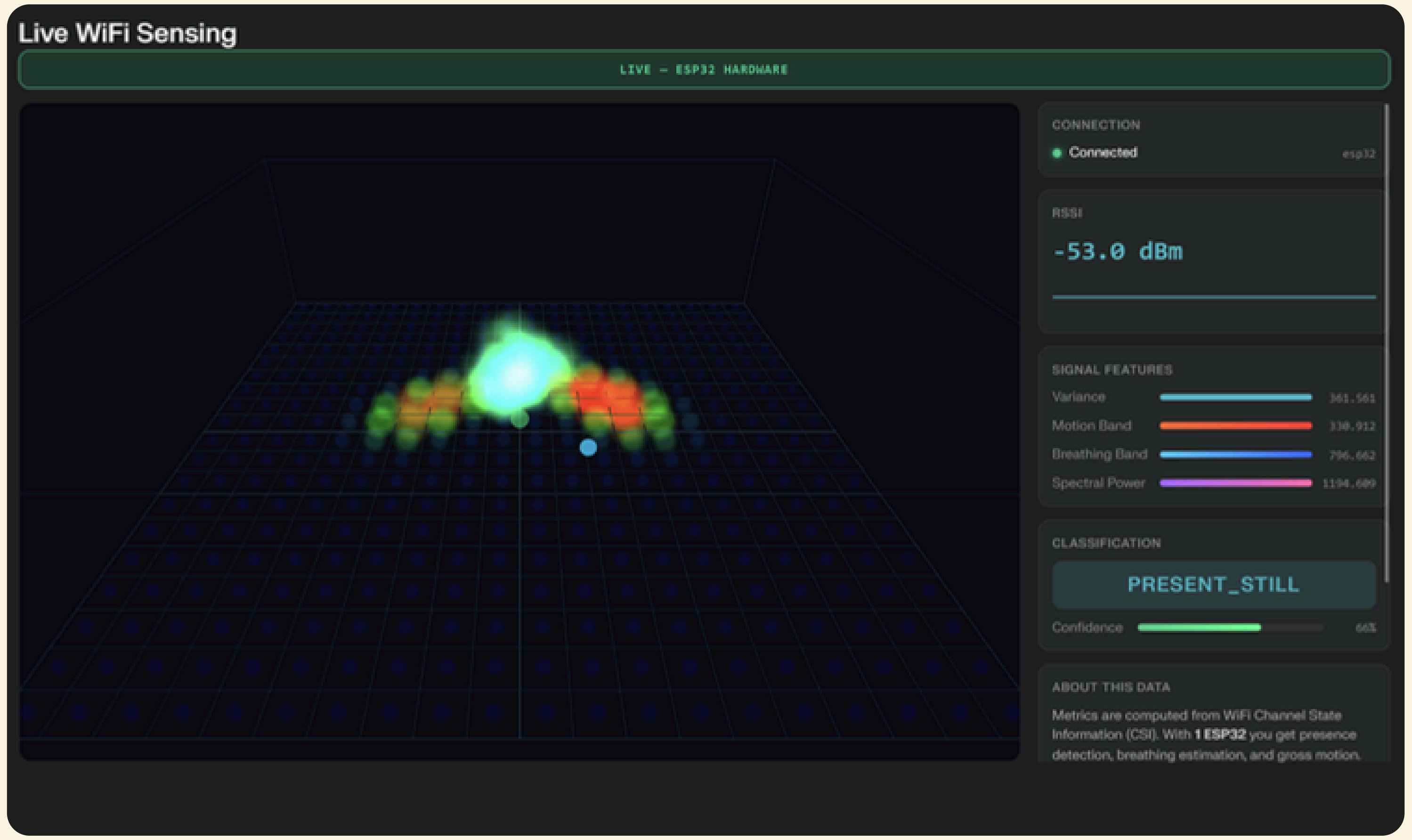
Alerts Sent
Caregivers receive real-time alerts

Prototype & Interface

Images of the Product:







Validation

As part of our validation and market research, we visited the **Punya Mitram Old Age home**, where we presented our solution to the management and caregivers. Their feedback indicated that the system would be highly useful, as it enables more effective monitoring while reducing the need for constant manual supervision.

Their insights helped us understand real-world challenges from a caregiver's perspective, allowing us to further refine the product. They also suggested additional features such as integrating **Blood Pressure (BP) monitoring**, which provides valuable direction for future enhancements.

Market Landscape

Category	Market Size (People)	Revenue Estimate	Reasoning
TAM (Total Addressable Market)	~3 crore	~₹3,000 crore	Paying urban + semi-urban elderly (~20–25% of 14 crore)
SAM (Serviceable Available Market)	~50–70 lakh	~₹500 – ₹1,000 crore	Elderly needing monitoring (urban focus)
SOM (Serviceable Obtainable Market)	~50,000 – 2 lakh (early adopters)	~₹5 – ₹20 crore	Initial realistic startup capture(5 - 10 years)

Key Market Trends

- Rapid growth in elderly population in India (~19.4 crore by 2031)
- Increasing nuclear families → elderly living alone
- Rising demand for remote healthcare monitoring
- Shift toward privacy-first technologies (less camera usage)
- Growth of IoT + AI in smart home environments

Market Potential

- ~14 crore elderly currently in India
- ~70% without continuous monitoring
- Realistic paying segment (TAM): ~3 crore people
- Serviceable segment (SAM): ~50–70 lakh people
- Initial target: Urban elderly living alone
- Early adopters (SOM): ~50,000 – 2 lakh users

Market & Competition

Competitors

Cameras → Privacy issues

Wearables → User dependency

Radar → Expensive

IoT sensors → Limited accuracy

Our Advantages

Privacy-first (no cameras)

WiFi-based (low cost)

Works through walls

Scalable

Feature	Cameras	Wearables	Radar	Our Product
Privacy	NO	YES	YES	YES
User Effort	YES	NO	YES	YES
Cost	MEDIUM	MEDIUM	NO	YES
Continuous	NO	NO	YES	YES
Through Walls	NO	NO	YES	YES

Business Model Canvas

Key Partners
- ESP32 hardware suppliers
- Healthcare providers
- Smart home companies
- Government agencies

Key Activities
- Hardware development
- AI signal processing
- Testing & calibration
- Product deployment

Key Resources
- ESP32 modules
- AI/ML expertise
- Software/backend
- Core team

Value Proposition
- Privacy-first monitoring
- No cameras, no wearables
- Detects breathing, motion, presence
- Works through walls & darkness
- Low-cost & scalable

Customer Segments
- Elderly individuals (Primary)
- Families & care homes
- Hospitals
- Disaster response

Channels
- Online sales
- Hospitals & care centers
- Smart home partnerships

Customer Relationships
- Subscription-based
- App dashboard
- Real-time alerts

Cost Structure
- Hardware
- AI/software
- Cloud
- Operations

Revenue Streams
- Device sales (₹2–3K)
- Subscription (₹199–499/month)
- B2B & Govt contracts

Revenue Model

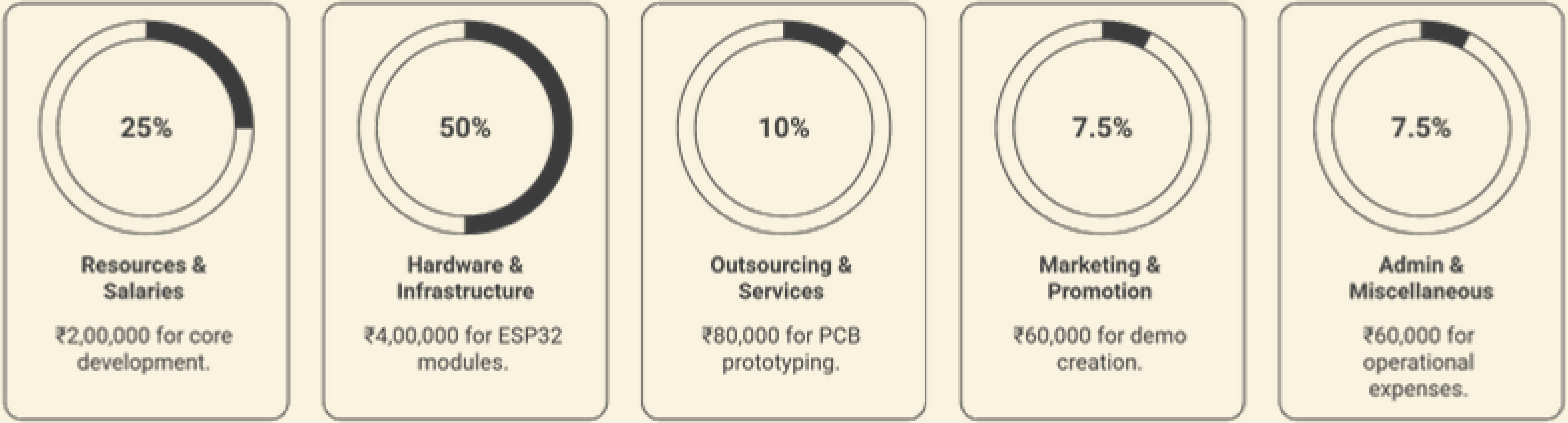
The pricing model for SPARS is designed to be affordable and scalable. The hardware device is priced between **₹2000 to ₹3000** as a one-time cost, making it accessible for households and institutions. Along with this, a subscription model of **₹199 to ₹299** per 6 months per room provides continuous monitoring, alerts, and dashboard access, ensuring ongoing value for users.

The revenue model combines one-time device sales with recurring subscription income as the main driver. Additional revenue comes from B2B bulk deployments in hospitals and care homes, along with government and enterprise contracts. This structure ensures steady growth and long-term profitability.

Year	Users	Revenue
Year 1	500	₹10–15 Lakhs
Year 2	3,000	₹60–90 Lakhs
Year 3	10,000	₹3+ Crore

Recurring subscription revenue ensures long-term scalability and profitability.

Budget



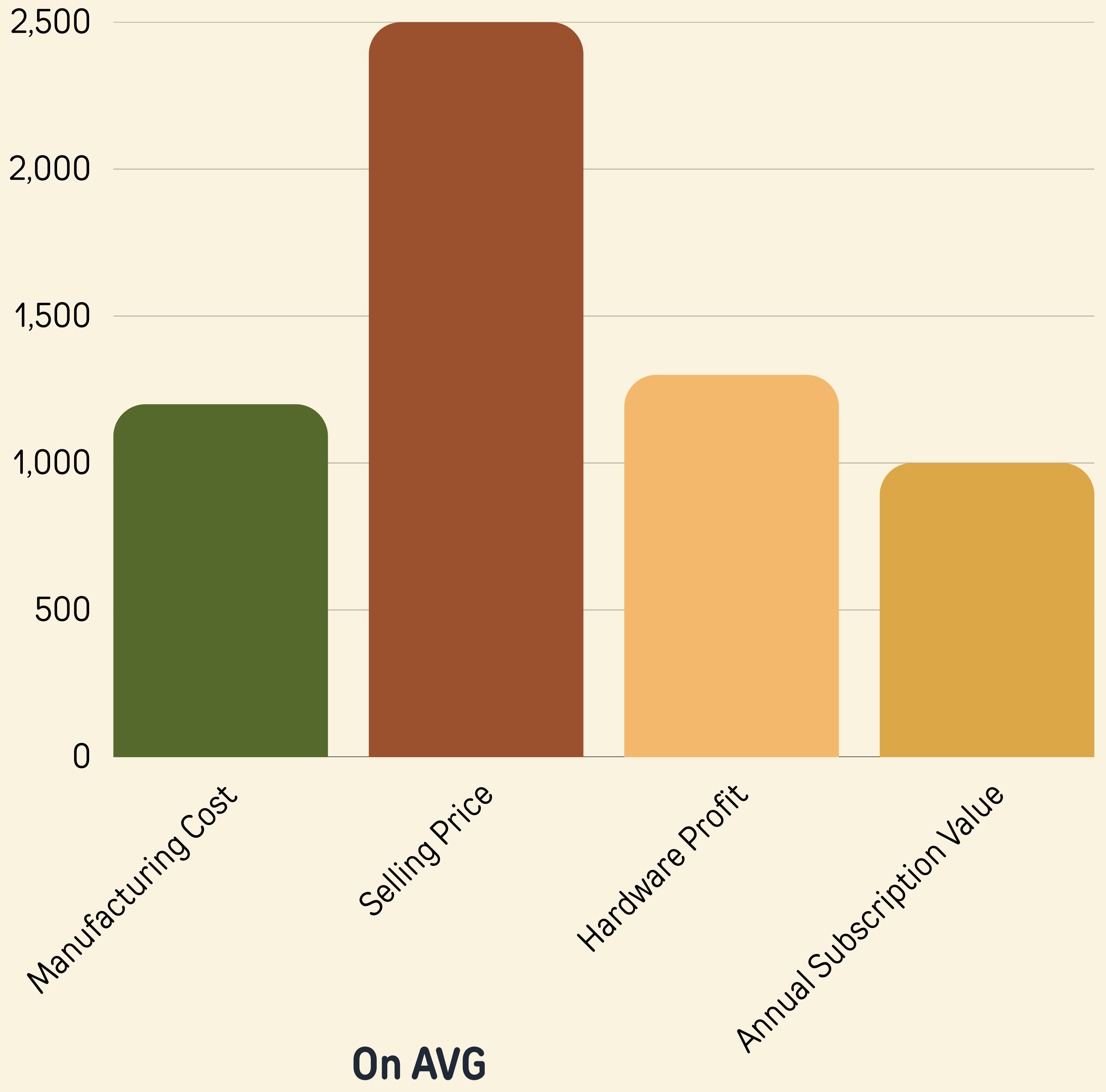
The total estimated budget for **SPARS** focuses on building a cost-efficient and scalable system by leveraging low-cost hardware and optimized development resources. The budget is distributed across key areas required to develop, test, and deploy the product effectively. This budget supports prototype development and an initial batch of around **300–400** units for testing and pilot deployment, with an approximate manufacturing cost of **₹1,000–₹1,500** per unit.

Once validated, the system can be scaled for larger production at an even lower per-unit cost, improving overall margins and enabling wider adoption.

Budget

SPARS requires an initial investment of **₹8 lakhs** to develop the system and produce an initial batch of approximately **300–400 units** for pilot deployment. Each unit has a manufacturing cost of **₹1000 to ₹1500** and is priced between **₹2000 to ₹3000 per unit**, ensuring a healthy hardware margin from the beginning. This allows early recovery of costs through device sales while validating the product in real-world environments.

In addition to hardware revenue, SPARS follows a subscription model priced at **₹199 to ₹299** per 6 months per room, generating consistent recurring income. This dual revenue approach by combining upfront device margins with ongoing subscription revenue which enables faster break-even and long-term profitability. Based on this model, the project is expected to break even at approximately **40–45 users** taking 2 rooms on average for each user, making it achievable within the early growth phase.



Customer Segment

1

Primary Users

- Elderly individuals living alone
- Families monitoring elderly members

2

Secondary Users

- Care homes & assisted living centers
- Hospitals (non-invasive monitoring)

3

Future Expansion

- Disaster rescue teams
- Smart home ecosystems

Distribution of Product / Service

- Direct-to-consumer: Online sales (website, marketplaces)
- B2B partnerships: Care homes, hospitals
- Institutional tie-ups: Government & smart city initiatives

SWOC

Strengths

- Privacy-first (no cameras)
- No wearable dependency
- Low-cost hardware (ESP32-based)
- Works through walls & low visibility
- Multi-industry scalability

Weaknesses

- Early-stage prototype (PoC level)
- Accuracy depends on environment
- Signal interference in crowded spaces
- Requires calibration for different setups

Challenges

- Competition from established technologies (radar, IoT, cameras)
- Need for high accuracy in real-world deployment
- User trust & adoption for new technology

Opportunities

- Growing elderly population & healthcare demand
- Expansion into smart homes & smart cities
- Disaster rescue & defense applications
- Increasing demand for privacy-focused tech

Socio-Economic Impact

Impact on Society
SPARS enables safe and independent living for elderly individuals through continuous, privacy-first monitoring. It detects emergencies like falls without cameras or wearables and works even in low-visibility or disaster environments.

Quantitative Impact
It reduces emergency response time by 40–60% and lowers dependency on caregivers, cutting healthcare costs. Its low cost and use of existing WiFi make it scalable across millions of homes.

Economic Impact
It supports caregiving systems, enables smart city integration, and creates opportunities in IoT, AI, and deployment services.

Qualitative Impact
SPARS restores dignity and privacy while giving families peace of mind. It transforms homes into intelligent, life-aware environments and promotes non-intrusive healthcare solutions.

Demonstration Link

[Demo Link]

THANK YOU